

ERP SaaS Module Dependency Map

Version ۲٫۰

Status: Architectural Decision Document

State: Consolidated Dependency Definition

Project: HamarehERP SaaS Platform

۱. Purpose

این سند روابط وابستگی بین Module های HamarehERP را مشخص می کند.
اهداف:

- جلوگیری از Dependency اشتباه
- جلوگیری از Circular Dependency
- تعیین ترتیب توسعه
- مشخص کردن جهت ارتباطات
- حفظ قابلیت مهاجرت آینده به Microservices

اصل پایه:

Dependency فقط از بالا به پایین است.

۲. Dependency Direction Model

ساختار نهایی:

SaaS Platform Layer

↓

Identity & Security Core

↓

ERP Foundation Layer



ERP Business Modules



Extensions & Integrations

۳. Dependency Rules

Rule ۱

Layer پایین‌تر اجازه Dependency به Layer بالاتر ندارد.

مثال غیرمجاز:

Identity



Accounting

Identity نباید چیزی درباره Accounting بداند.

Rule ۲

Core مستقل است.

Identity نباید وابسته باشد به:

- Inventory
- Sales
- Accounting
- Manufacturing

Rule ۳

Business Module ها فقط از Core و Foundation استفاده می کنند.

مثال:

مجاز:

Inventory



Organization



Identity

غیرمجاز:

Inventory



Sales



Purchase

۴. Global Dependency Graph

SaaS Platform

Tenant Management



|
Subscription & Billing

|

|

Payment Gateway

↓

Identity Core

Identity Management

Authorization

Scope Management

↓

ERP Foundation

Organization

Master Data

Workflow

Document Management



ERP Business Modules

Accounting

Inventory

Sales

Purchase

Manufacturing

HR

Project Management

Δ. SaaS Platform Dependencies

Tenant Management

Dependency:

None

Used By:

Identity

Subscription

Billing

Notification

ERP Foundation

Communication:

API •

Logical Reference •

Subscription & Billing

Depends On:

Tenant Management

Used By:

All Licensed Modules

Responsibility:

Subscription State •

Feature Availability •

License Check •

Payment Gateway

Depends On:

Subscription & Billing

Communication:

InvoiceCreated.v\

PaymentSucceeded.v\

9. Identity Core Dependencies

Identity Management

Depends On:

Tenant Management

Provides:

User Context

Authentication

JWT

Session

Used By:

All Modules

Authorization

Depends On:

Identity Management

Provides:

Role

Permission

Scope Validation

Used By:

All Modules

V. ERP Foundation Dependencies

Organization Management

Depends On:

Tenant Management

Identity Core

Used By:

All ERP Modules

Provides:

Company

Branch

Department Context

Master Data

Depends On:

Organization

Used By:
Accounting

Inventory

Sales

Purchase

Manufacturing

Workflow Engine

Depends On:
Identity

Organization

Used By:
All Modules requiring Approval

Document Management

Depends On:
Tenant

Identity

Used By:
All Modules

A. Business Module Dependencies

Accounting

Depends On:

Identity

Organization

Master Data

Document Management

Consumed By:

Sales

Purchase

Inventory

Manufacturing

Produces:

JournalCreated.v\

Inventory

Depends On:

Identity

Organization

Master Data

Document Management

Consumed By:

Sales

Purchase

Manufacturing

Produces:

InventoryChanged.v\

Sales

Depends On:

Identity

Organization

Master Data

Inventory

Accounting

Produces:

SalesConfirmed.v\

Purchase

Depends On:
Identity

Organization

Master Data

Inventory

Accounting

Produces:

PurchaseCompleted.v\

Manufacturing

Depends On:
Identity

Organization

Inventory

Accounting

Produces:

ProductionCompleted.v\

HR

Depends On:

Identity

Organization

No dependency on:

Accounting

Inventory

Sales

Project Management

Depends On:

Identity

Organization

No dependency on Business Modules.

۹. Forbidden Dependencies

موارد زیر ممنوع هستند:

Business To Business Direct Dependency

غیرمجاز:

Sales

↓

Inventory Database

Database Coupling

غیرمجاز:

Sales.orders

FK

Inventory.stock

Circular Dependency

غیرمجاز:

Sales

↓

Accounting

↓

Sales

۱۰. Communication Strategy

Query Operations

استفاده از:

API Contract

Service Interface

مثال:

Sales:

"آیا مشتری اعتبار دارد؟"

درخواست:

Accounting Credit Service

State Changes

استفاده از:

Domain Event

مثال:

SalesConfirmed.v۱



Inventory



Accounting

۱۱. Development Order

Phase •

Infrastructure

شامل:

- Repository
- Docker
- CI/CD
- Base Application

Phase ۱

SaaS Core

ترتیب:

۱. Tenant Management

۲. Identity

Authorization .۳

Subscription .۴

Billing .۵

Notification .۶

Audit .۷

Phase ۲

ERP Foundation

ترتیب:

Organization .۱

Master Data .۲

Document Management .۳

Workflow .۴

Phase ۳

Business Modules

ترتیب:

۱. Accounting

دلیل:

مرجع مالی تمام عملیات.

۲. Inventory

دلیل:

وابستگی عملیاتی بالا.

۳. Sales

۴. Purchase

۵. Manufacturing

۶. HR

V. Project Management

۱۲. Future Evolution Path

معماری فعلی:

Modular Monolith

مسیر آینده:

Modular Monolith



Service Oriented Architecture



Microservices

بدون بازطراحی کامل.

۱۳. Final Dependency Statement

قانون نهایی:

Platform



Core



Foundation



Business Modules



Extensions

هیچ توسعه‌ای خارج از این مسیر مجاز نیست.

